Case Study 8: Text Compression Using Character Runs
Complete Python Source Code

```
text = "aaabbccccd"
if text == "":
    encoded = ""
else:
    encoded = ""
    count = 1
    for i in range(1, len(text)):
        if text[i] == text[i - 1]:
            count += 1
        else:
            encoded += text[i - 1] + str(count)
            count = 1
    encoded += text[-1] + str(count)
if len(encoded) < len(text):
    chosen = encoded
else:
    chosen = text
print("Original text:", text)
print("Encoded text:", encoded)
print("Original length:", len(text))
print("Encoded length:", len(encoded))
print("Chosen representation:", chosen)
```